

PUSHKINSKIYE GORY, Russian Federation

WMO: 263590

Lat: 57.017N

Lon: 28.900E

Elev: 108

StdP: 100.03

Time Zone: 3.00 (E03)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.9	-18.7	-24.6	0.4	-21.4	-21.6	0.6	-18.1	7.1	-1.3	6.3	-0.7	1.3	20	0.522

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.8	28.2	20.4	26.4	19.3	24.6	18.3	21.2	26.6	20.2	24.9	19.2	23.5	2.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.3	14.3	23.8	18.4	13.5	22.9	17.5	12.7	21.8	62.1	26.9	58.4	25.0	55.1	23.4	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.8	5.2	4.5	-24.3	30.6	4.7	2.0	-27.7	32.0	-30.4	33.2	-33.1	34.3	-36.5	35.8		
			-24.5	22.8	4.6	1.0	-27.9	23.6	-30.6	24.2	-33.2	24.8	-36.5	25.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.4	-5.9	-5.0	-0.2	6.5	12.8	15.7	18.6	17.2	12.3	5.8	1.3	-2.6
	DBStd	9.62	6.87	6.38	4.77	4.15	4.28	3.42	3.33	3.28	3.44	4.03	4.26	5.50
	HDD10.0	2176	492	419	317	119	20	2	0	0	16	139	261	391
	HDD18.3	4436	750	653	575	354	179	93	38	62	182	389	510	649
	CDD10.0	875	0	0	0	14	106	172	266	222	86	9	0	0
	CDD18.3	94	0	0	0	0	7	13	46	26	2	0	0	0
	CDH23.3	676	0	0	0	1	58	99	316	194	8	0	0	0
	CDH26.7	134	0	0	0	0	9	13	67	45	0	0	0	0
Wind	WSAvg	2.3	2.6	2.4	2.5	2.4	2.1	1.9	1.7	1.9	2.0	2.3	2.6	2.7
Precipitation	PrecAvg	693	52	40	38	39	61	84	72	79	61	66	57	44
	PrecMax	921	82	68	85	77	116	178	171	153	125	132	87	69
	PrecMin	514	16	10	11	2	14	9	13	1	16	13	0	18
	PrecStd	97	18	16	19	22	25	45	42	44	29	30	21	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.7	5.6	14.2	22.2	27.9	28.4	30.8	31.0	24.4	17.0	10.5	8.7
		MCWB	5.6	3.7	7.9	11.3	19.4	20.1	21.7	22.4	17.9	12.9	9.1	7.7
	2%	DB	4.3	4.0	10.5	18.7	24.7	26.1	28.5	27.6	21.7	14.1	8.8	6.8
		MCWB	3.4	2.6	5.2	10.4	16.7	18.4	20.9	20.0	16.6	11.9	7.6	5.9
	5%	DB	2.8	2.9	7.7	16.1	22.6	24.1	26.7	25.4	19.6	12.6	7.5	5.0
		MCWB	1.9	1.8	4.1	9.2	15.2	17.4	20.0	19.0	15.6	10.9	6.5	4.2
	10%	DB	1.7	1.8	5.6	13.8	20.4	22.3	24.9	23.4	17.7	11.4	6.4	3.5
		MCWB	1.0	0.9	3.0	7.8	14.0	16.6	19.0	18.1	14.5	10.0	5.6	2.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.9	4.4	8.2	12.4	19.9	21.1	22.5	22.8	19.1	14.0	9.7	7.9
		MCDB	6.8	5.3	13.7	19.0	26.9	26.7	28.6	29.7	22.0	15.5	10.3	8.6
	2%	WB	3.5	2.7	5.8	11.1	17.9	19.5	21.5	21.1	17.3	12.3	7.8	6.0
		MCDB	4.2	3.5	9.1	16.9	23.1	24.3	27.1	26.1	20.7	13.5	8.5	6.6
	5%	WB	2.1	2.0	4.5	10.0	16.3	18.3	20.5	19.7	16.1	11.2	6.6	4.3
		MCDB	2.7	2.8	7.0	15.5	20.8	22.6	25.5	23.8	18.9	12.4	7.3	4.9
	10%	WB	1.1	1.1	3.4	8.5	14.9	17.2	19.7	18.6	14.9	10.2	5.6	2.7
		MCDB	1.7	1.8	5.2	13.0	19.4	21.4	24.0	22.6	17.4	11.3	6.3	3.4
Mean Daily Temperature Range	5% DB	MDBR	4.4	5.0	6.7	9.2	10.0	9.2	8.8	8.8	7.6	5.3	3.6	3.7
		MCDBR	3.5	4.3	9.2	13.0	12.7	11.5	11.2	11.8	10.3	6.9	4.8	3.7
		MCWBR	3.4	3.4	5.9	6.7	6.1	5.8	5.2	5.5	5.9	4.9	4.2	3.4
	5% WB	MCDBR	3.4	3.7	8.0	11.4	11.1	9.9	10.0	10.2	9.2	5.9	4.6	3.7
		MCWBR	3.4	3.3	5.6	6.5	6.0	5.7	5.1	5.6	5.7	4.7	4.2	3.5
Clear-Sky Solar Irradiance	taub	0.281	0.315	0.343	0.386	0.379	0.373	0.399	0.401	0.366	0.342	0.322	0.279	
	taud	2.137	2.126	2.169	2.253	2.350	2.389	2.334	2.323	2.398	2.386	2.239	2.151	
	Ebn at Noon	576	712	800	824	858	868	833	802	779	693	525	471	
	Edh at Noon	55	86	108	117	113	110	114	108	87	68	52	42	
All-Sky Solar Radiation	RadAvg	0.47	1.28	2.66	3.94	5.24	5.52	5.30	4.23	2.70	1.21	0.41	0.26	
	RadStd	0.11	0.23	0.43	0.53	0.53	0.39	0.54	0.37	0.27	0.22	0.06	0.07	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-153	N/A	N/A	N/A

Nomenclature: See separate page